

KOMAL NAKRANI / PROFESSIONAL FIELD GUIDES

Forward Deployed Engineering

From Ambiguous Workflow to Production Outcome



Komal Nakrani

SCREEN-FIRST
DESIGN PROOF

01 A Komal Nakrani field guide

A screen-first system for reading consequential technical work with less friction and stronger orientation.

This is a private design proof - not the published edition.

ORIENT

Route maps establish the territory and the next coordinate.

READ

Evidence rails hold context without interrupting the main argument.

DECIDE

Signal orange appears only when judgment changes the route.





FIELD LOG / 01

Forward Deployed Engineering

From Ambiguous Workflow to
Production Outcome

Komal Nakrani

First edition, version 1.0.0 (2026)

Edition and authorship record

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Design proof - private review artifact. This file does not replace the published first edition.

The fictional Orchid Equipment Services and all constructed satellite scenarios are not real customer or employer outcomes.

AUTHOR

Komal Nakrani

PROOF FORMAT

7 x 10 in / screen-first

PUBLICATION STATE

Published edition unchanged

VISUAL SYSTEM

Field Expedition Log

PROFESSIONAL BOUNDARY

This book prepares evidence, supports bounded implementation, recommends options, and preserves escalation. Formal risk decisions remain with qualified specialists and designated organizational authorities.

How to use this field log

Orient. Inspect. Decide. Practice.

- 01 Read Chapters 1-5 before committing scope.
- 02 Use Chapters 6-10 to create the design dossier.
- 03 Run the companion while studying Chapters 11-14.
- 04 Use Chapters 15-17 for readiness, stabilization, and ownership.
- 05 Use Chapters 18-19 only after bounded field evidence exists.

FIELD PRACTICE

Complete the learning packs and deterministic companion exercises. They are professional education, not a live certification bank.

Read the field marks

01

EVIDENCE

a fact, observation, measurement, or source whose limits remain visible.

02

DECISION

a consequential choice that changes scope, exposure, ownership, or next action.

03

WARNING

a condition that can invalidate the current path.

04

COORDINATE

your current position in the deployment journey.

05

HANDOFF

the artifact and owner required before the route can continue.

Expedition route

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Expedition route

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Design the Deployment

Convert the selected workflow into explicit boundaries that reviewers can challenge.

Boxes alone are not architecture. The deployment must state which system holds which responsibility, which proposition crosses each interface, how identities and regions are enforced, where uncertain model behavior belongs, and which control evidence supports formal decisions.

Draw the Real System Boundary

Forward deployed architecture begins where boxes and arrows stop.

An architecture diagram can be accurate and still conceal the deployment.

It may show a user, an application, several services, and databases connected by arrows. Yet it may not say who operates, authorizes, detects, reconciles, or recovers.

Draw the boundary the outcome needs

Chapter 5 selected one Orchid path: a Region West cohort and equipment family moving from confirmed equipment match through visible evidence, bounded suggestion, deterministic eligibility, qualified approval, read-only inventory evidence, audit, fallback, signals, and support.

Do not begin by selecting cloud services. Begin with responsibilities and constraints.

? Which actors and systems participate from trigger to supported resolution or escalation?

? Which customer systems remain authoritative for which claims?

? Which failure can propagate, and where can it be detected or contained?

? Which team operates, changes, supports, and approves each responsibility?

Six boundaries, not one box

A structural diagram does not establish the runtime, data, trust, failure, or responsibility boundary.

Each view answers a different operational question. Keeping them separate prevents a clean diagram from hiding a fragile deployment.

**A clean structural picture
can still conceal a brittle
operating system.**

01

STRUCTURAL

Which components and relationships exist?

02

RUNTIME

Where does code execute, restart, scale, and persist?

03

DATA

Which proposition crosses, and who remains authoritative?

04

TRUST

Where must identity and authorization be evaluated?

05

FAILURE

What can propagate, diverge, or require reconciliation?

06

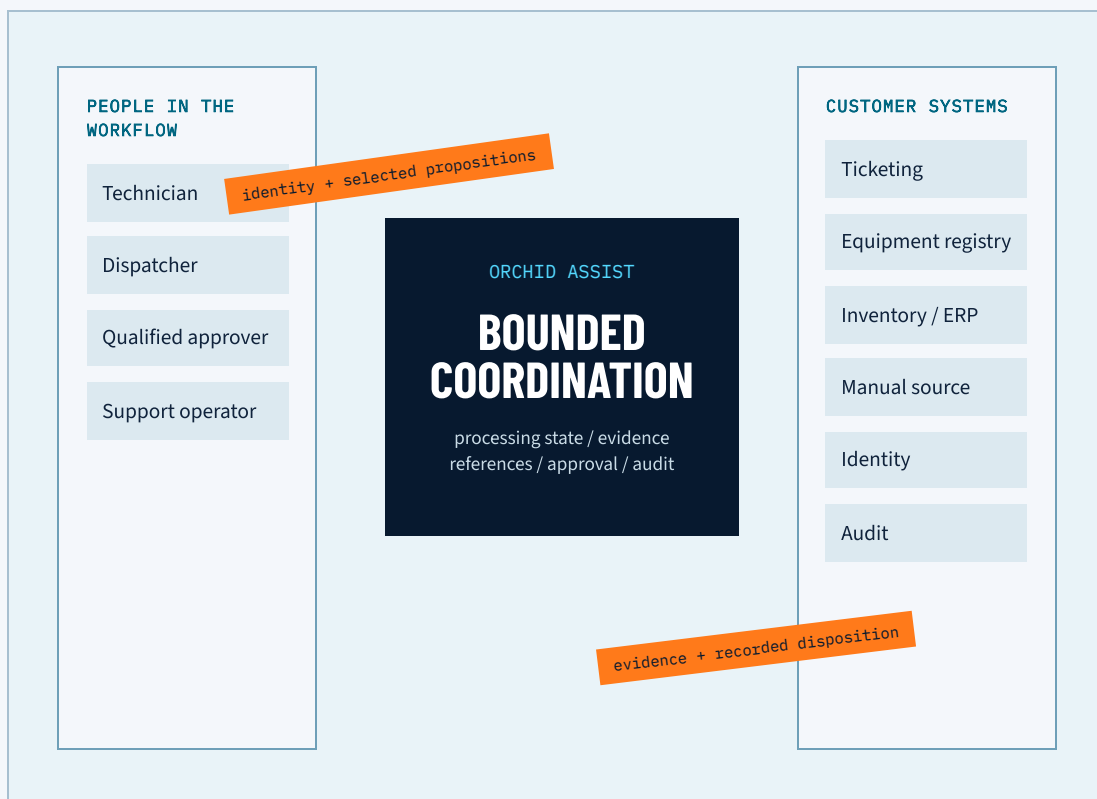
RESPONSIBILITY

Who builds, operates, supports, changes, and decides?

Orchid Assist is bounded coordination

Orchid Assist coordinates the selected workflow. It does not become the owner of every customer record or decision merely because it reads or displays them.

A system context becomes useful only when every external relationship names the proposition, identity, limitation, and owner.



A boundary is a decision surface

PICTURE

A clean architecture picture

Components

Connections

Nominal data flow

Technology names

OPERATING TRUTH

A deployed system boundary

Proposition ownership

Identity and authority

Failure and reconciliation

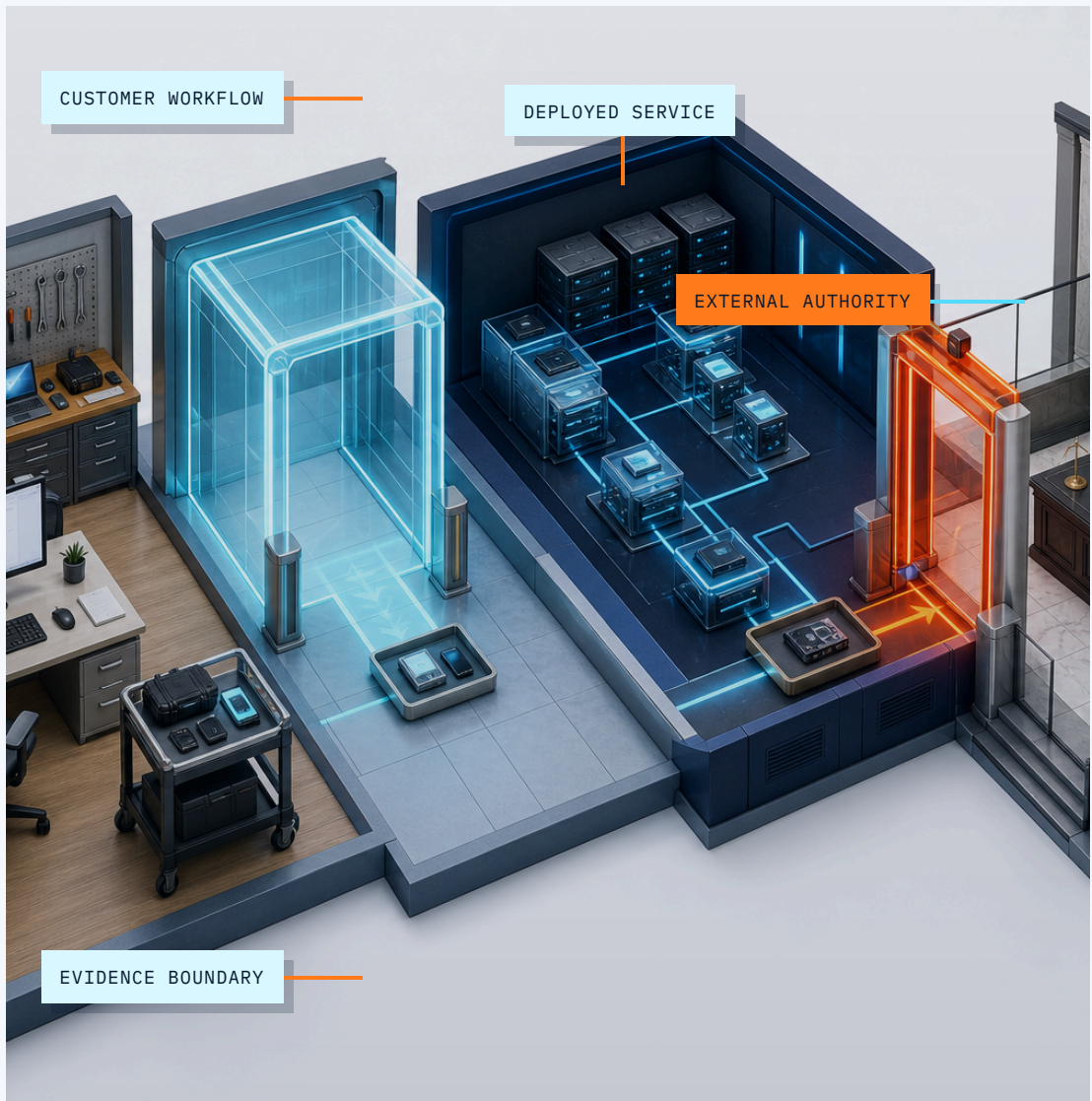
Operations and support

Change and recovery

DECISION

If a reviewer cannot challenge ownership, trust, and failure, the picture is not yet a deployment boundary.

The real system boundary is layered



Illustrative field model. The visual separates operational zones; the surrounding labels carry the exact boundary meaning.

Read the figure as evidence, not decoration

The deployed service can coordinate a workflow without inheriting authority over the customer systems it touches.

Every crossing should be read as a question: which proposition moves, which identity acts, which owner supplies evidence, and what happens when the crossing fails?

Every crossing is a question before it becomes an interface.

01

Workflow remains the customer context, not a decorative backdrop.

02

Evidence is selected and bounded before it crosses into the service.

03

The deployed service owns its processing state, not every underlying proposition.

04

Formal authority stays external unless explicitly delegated.

Decision checkpoint

06 / GATE

**Can every material relationship
answer who does what to whom, why,
with which identity and data, and
under whose authority?**

**REQUIRED
EVIDENCE**

Relationship sentence

Proposition owner

Identity and scope

Failure behavior

Support and decision owner

DISPOSITION

If any one is missing, keep the boundary provisional and block irreversible implementation.

Artifact: access-purpose chain

```
01    discovery question
02      -> minimum credible evidence
03      -> minimum data or access
04      -> handling constraint
05      -> owner and approval
06      -> expiry
07      -> decision enabled
```

Questions are not decoration before access. They are the basis for access.

Purpose-bound evidence makes it easier to record what a finding does and does not support.

MINIMUM ACCESS / NAMED DECISION

01

PURPOSE BOUND

The question determines which evidence is legitimate to request.

02

OWNER NAMED

Approval and handling remain attached to an accountable role.

03

EXPIRY RECORDED

Access ends when the named decision no longer needs it.

RESULT

ONE BOUNDED DECISION BECOMES LEGITIMATE TO MAKE.

Walk the failure before it walks you

Choose one material dependency and remove it. Trace the user effect, detection path, containment, recovery, reconciliation, and owner.

01

Remove one dependency: registry, inventory, identity, audit, telemetry, or configuration.

02

State whether the path blocks, falls back, degrades, queues, or continues.

03

Name what evidence proves the resulting state.

04

Name who may decide, who recovers, and who communicates.

PASS WHEN: the route ends with a named owner and a reconciled state, not an optimistic retry.

Coordinate 06 complete

06

You can now separate structural, runtime, data, trust, failure, and responsibility boundaries.

The next coordinate makes the propositions crossing those boundaries explicit: contracts, time, intent, finality, quality, lineage, and reconciliation.

- Boundary views are distinct
- Every arrow has a relationship sentence
- Failure ends with ownership
- Desired and operational state are not confused



Field template: boundary record

BOUNDARY ID Stable reference used by prose, risks, tests, signals, and runbooks.

PURPOSE The outcome, guardrail, or scope decision that requires this boundary.

CROSSING Actor, proposition, identity, operation, owner, and direction.

FAILURE Propagation, detection, containment, correction, and reconciliation.

AUTHORITY Who approves, rejects, accepts consequence, or grants exception.

EVIDENCE What shows the boundary is designed, implemented, verified, and operable.

Sources, figures, and accessible reading

Read every figure with its caption and long description. The figure supports the explanation; it does not replace evidence or authority.

Source records remain attached to the claims they support. Framework mappings are not certification, and professional decisions stay with qualified specialists and designated organizational authorities.

FIGURE

PNG asset + selectable labels + visible caption + descriptive alternative.

CLAIM

Exact statement + source IDs + limitations + chapter placement.

EDITION

Version + date + errata + canonical route + download identity.

TRACEABILITY IS PART OF THE READING EXPERIENCE.



AUTHOR / ORIGINAL PROFESSIONAL EDUCATION

Komal Nakrani

Komal Nakrani is the author of Forward Deployed Engineering.

This edition is built to help practicing and aspiring technical professionals reason about real work, preserve evidence boundaries, and make bounded decisions.

The separate Abhyaas product independently certifies competence; this book teaches and prepares evidence but is not a certification exam.

Komal Nakrani

Leave the system more legible than you found it.

A deployment is complete only when its outcome, evidence, limitations, operations, ownership, and next decisions remain visible without the person who first carried it across the boundary.

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